

1.

	MAC	DAC	RBAC	ABAC
Control Authority	System administrator or central authority	Owner of the resource or data	Administrator defined roles	Admin defined policies based on attributes
Basis of Decision	Security classifications and labeling	Ownership and user specified permissions	Roles of users in the system	Attributes (activity, environment, resource, and user)
Flexibility	Fixed roles. Low	User delegates access. High	Role based on assignment. Medium	Role context aware and dynamic. High
Granularity	Access is strictly regulated.	Per user, per file	Determined by the role	Examines a variety of attributes
Ease of Management	Challenging to scale	Small systems are simple, but huge systems are more difficult.	Having clear responsibilities and permissions makes things easier.	Although difficult to set up, once specified, it is effective.
User Involvement	User access rights cannot be modified	Users control who can access their resources	Roles, not direct entitlements, are assigned to users	Entitlements based off policy determines access
Scalability	Poor	Poor	Good	Excellent
Common Use Case	Government role	Small companies	HR and financial systems	Cloud services

2. Sender (S):

1. Create a message digest ($H(M)$) by hashing the message (M) using a secure hash method (SHA-256).

2. Create a digital signature by encrypting the hash using the sender's private key. Since the signature could only have been created with S's private key, non-repudiation is guaranteed.
3. Since only R can decode the original message (M), encrypt it with the recipient's public key to maintain confidentiality.
4. Provide the recipient the digital signature along with the encrypted communication.

Receiver (R):

1. Use the private key to decrypt the encrypted message. Since only R has access to the original message, confidentiality is guaranteed in this step.
2. To recover the original hash, decrypt the digital signature with the sender's public key. This guarantees nonrepudiation and confirms that S did, in fact, sign the message.
3. To create a fresh digest, hash the decrypted message (M) once more using the same hashing technique.
4. Compare the recently calculated hash with the signature's decrypted hash. If they match, integrity is guaranteed and the message remains unaltered. The message might have been changed in transit if they don't match.

3. In firewall security policy, blacklisting and whitelisting are two conflicting strategies. By default, blacklisting permits all traffic, with the exception of sources or destinations that are specifically blacklisted. Particularly in dynamic situations where new apps or services are regularly deployed, it is adaptable and simpler to implement initially. However, because only known bad actors are prevented, it is reactive and exposes systems to unknown or zero day threats. In contrast, whitelisting only allows sources or services that have been specifically approved and by default rejects all traffic. By reducing exposure to harmful or unauthorized traffic, this approach offers a better level of security; but, it is more restricted and challenging to maintain, particularly in big or dynamic environments. If legal traffic is not appropriately permitted, it may potentially result in operational interruptions. Whitelisting is more secure but involves careful preparation and continuous administration, however blacklisting is more convenient but riskier.

4. By recognizing, categorizing, and monitoring sensitive data across endpoints, networks, and storage, an efficient Data Loss Prevention (DLP) strategy safeguards it. By implementing controls like encryption, blocking, or quarantining data when necessary, it employs established policies to stop unwanted access or transmission. To identify and address such breaches, the strategy should make use of user behavior analysis, context aware monitoring, and content discovery technologies. Along with regular staff training and incident response procedures, it must also comply with legal obligations (such as GDPR and HIPAA). Data security without interfering with lawful business operations is the aim.